

Purchase Pattern Analytics using Market Basket Analysis

1. Introduction

Purchase Pattern Analytics aims to understand customer buying behavior by analyzing historical retail transaction data. This project leverages Market Basket Analysis (MBA) techniques to discover meaningful associations between products frequently purchased together. The insights derived help businesses improve cross-selling strategies, product bundling, inventory planning, and overall customer experience.

2. Problem Statement

Retail businesses generate large volumes of transactional data, but identifying hidden relationships between products remains a challenge. The objective of this project is to apply Market Basket Analysis using the Apriori algorithm on transaction-level data to uncover frequent itemsets and strong association rules. The project also focuses on visualizing purchase trends and analytical results using Power BI to enable data-driven decision-making.

3. Dataset Description

The dataset consists of retail transaction records where each row represents an item purchased as part of a bill. The structure of the dataset is as follows:

- BillNo – Unique transaction identifier
- Itemname – Product name
- Quantity – Number of units purchased
- Present Date – Date and time of transaction
- Price – Unit price of the item
- CustomerID – Customer identifier
- Country – Country of transaction

The dataset contains over 5,22,063 lakh rows, 19,559 distinct transactions, and 4,006 unique items.

4. Data Import and Initial Exploration

The transactional dataset was imported into Python using the **Pandas** library to enable preprocessing and analytical operations. Supporting libraries such as **NumPy**, **Matplotlib**, **Seaborn**, and **mlxtend** were imported to facilitate numerical processing, visualization, and Market Basket Analysis.

Initial exploration was conducted to:

- Review the number of records and columns
- Understand the nature of each attribute
- Identify missing values and data inconsistencies
- Examine basic statistical properties of numerical fields

This exploration revealed that:

- The dataset contained a large volume of transactional records, with each row representing an item purchased within a bill
- The CustomerID column had a significant number of missing values
- Date-related fields were not in a suitable format for time-based analysis
- Certain data quality issues needed to be addressed before visualization and algorithm implementation

These observations helped define the required data cleaning and transformation steps.

5. Data Cleaning and Transformation

Data cleaning was carried out to improve data quality, ensure consistency, and make the dataset suitable for both visualization and Market Basket Analysis.

5.1 Date Standardization

The transaction date column was standardized into a proper date format. This step was necessary to enable accurate time-based analysis, such as identifying purchase trends across months and years in Power BI.

5.2 CustomerID Standardization

The CustomerID column required multiple transformations:

- Missing CustomerID values were replaced with the label "**Guest**" to retain transactions without registered customers
- The column was converted from float to string datatype
- Trailing decimal values (.0) present in customer identifiers due to numeric representation during data import were removed to maintain clean and consistent identifiers.

This step resolved datatype inconsistencies and prevented errors during data export and Power BI integration.

5.3 Handling Missing Item Names

Records with missing Item names were removed. Since product identification is essential for purchase pattern and association analysis, retaining such records would not contribute meaningful insights.

5.4 Filtering Invalid Transactions

To ensure only valid retail transactions were analyzed, transactions with **non-positive values** were removed from the dataset.

- Transactions where **quantity was less than or equal to zero**
- Transactions where **price was less than or equal to zero**

Such records typically represent returns, cancellations, free items, or data entry errors. Retaining them would distort purchase behavior analysis and negatively impact both descriptive insights and association rule mining. Therefore, only transactions with **positive quantity and positive price values** were retained for further analysis.

5.5 Final Data Validation and Export

After completing all cleaning steps, the dataset was reviewed to confirm:

- Correct data formats
- Removal of invalid records
- Overall data consistency

The cleaned dataset was then exported and used as:

- The input for **purchase trend and descriptive analysis in Power BI**
- The foundation for **Market Basket Analysis and association rule mining in Python**

6. Purchase Trend and Descriptive Analysis using Power BI

Power BI was used to visualize overall purchasing behavior and trends in the data. Key analyses included:

- Products with highest purchase frequency
- Purchase trends over time using date hierarchy (Year, Quarter, Month, Day)
- Total number of bills and unique products
- Total Products by Quantity sold
- Average number of items purchased per bill
- Country-based filtering for comparative analysis

These visualizations provided a comprehensive, high-level understanding of customer purchasing patterns and transactional behavior. This descriptive analysis helped validate data quality, identify key trends, and set the context for subsequent Market Basket Analysis and association rule mining.

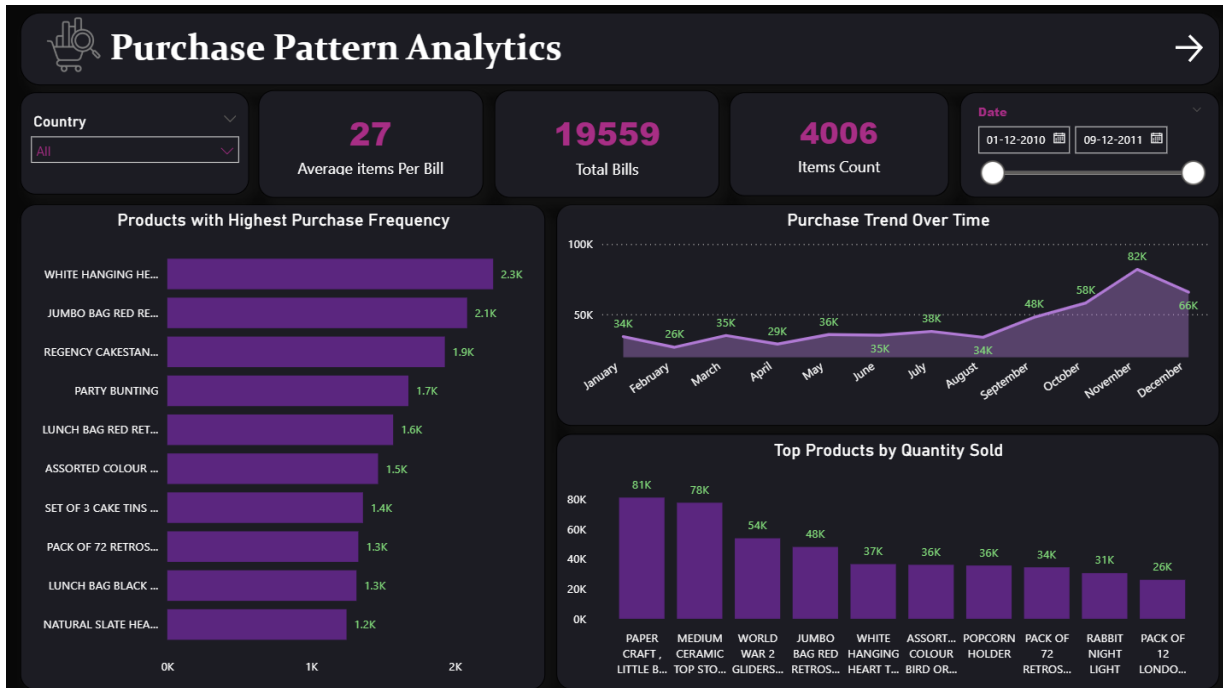


Fig: Purchase Trends & Distributions

7. Basket Creation for Market Basket Analysis

To perform Market Basket Analysis, transactions were grouped by BillNo, where each bill represents a single customer purchase session. Item names within each bill were aggregated to form transaction baskets. This transformation converted transactional data into a suitable format for association rule mining.

8. Transaction Encoding

The TransactionEncoder from the mlxtend library was used to convert item baskets into a **boolean transaction matrix**, where each row represents a transaction and each column represents an item. The matrix indicates the **presence or absence** of items within each transaction.

9. Apriori Algorithm Implementation

The Apriori algorithm was applied to the encoded transaction matrix to identify frequently occurring combinations of products. A minimum support threshold of **2%** was selected to filter out infrequent item combinations while retaining meaningful purchasing patterns.

The algorithm generated frequent itemsets of varying sizes in a single execution. These itemsets were then categorized based on their size to analyze:

- **Single-item itemsets**, representing frequently purchased individual products
- **Two-item itemsets**, representing commonly co-purchased product pairs
- **Three-item itemsets**, representing frequently purchased product triplets

To identify the most significant combinations, itemsets were grouped by their size and ranked in descending order of support. This enabled focused analysis of the most frequent product pairs and triples observed in customer transactions.

10. Association Rule Mining

After identifying frequent itemset, association rule mining was performed to uncover meaningful relationships between products purchased together. The objective of this step was to determine how strongly the presence of one product influences the purchase of another.

Association rules were generated using the following evaluation metrics:

- Support measures how frequently an itemset appears in the dataset
- Confidence indicates the likelihood of purchasing one item given another
- Lift evaluates the strength of association beyond random chance

Initially, rules with a lift value greater than 1 were generated and ranked to identify positive product relationships. To focus on the most actionable insights, rules with **high lift and high confidence** were retained, ensuring that only strong and reliable product associations were considered.

For improved readability and visualization, product combinations within the rules were converted into clear, human-readable text formats. The finalized frequent itemsets and association rules were then exported for visualization and analysis in Power BI.

11. Visualization of Results

Power BI dashboards were developed to present the results of the Market Basket Analysis in a clear and intuitive manner. The purpose of these visualizations was to convert complex analytical outputs into easily interpretable insights:

The dashboards included:

- Top frequent itemsets
- Frequent item pairs by support
- Frequent triple itemsets
- Strong association rules ranked by lift and confidence

These visualizations enabled quick identification of key product associations and purchase patterns. Presenting the results in an interactive visual format made it easier to analyze and interpret the outcomes of the Apriori algorithm and association rule mining.

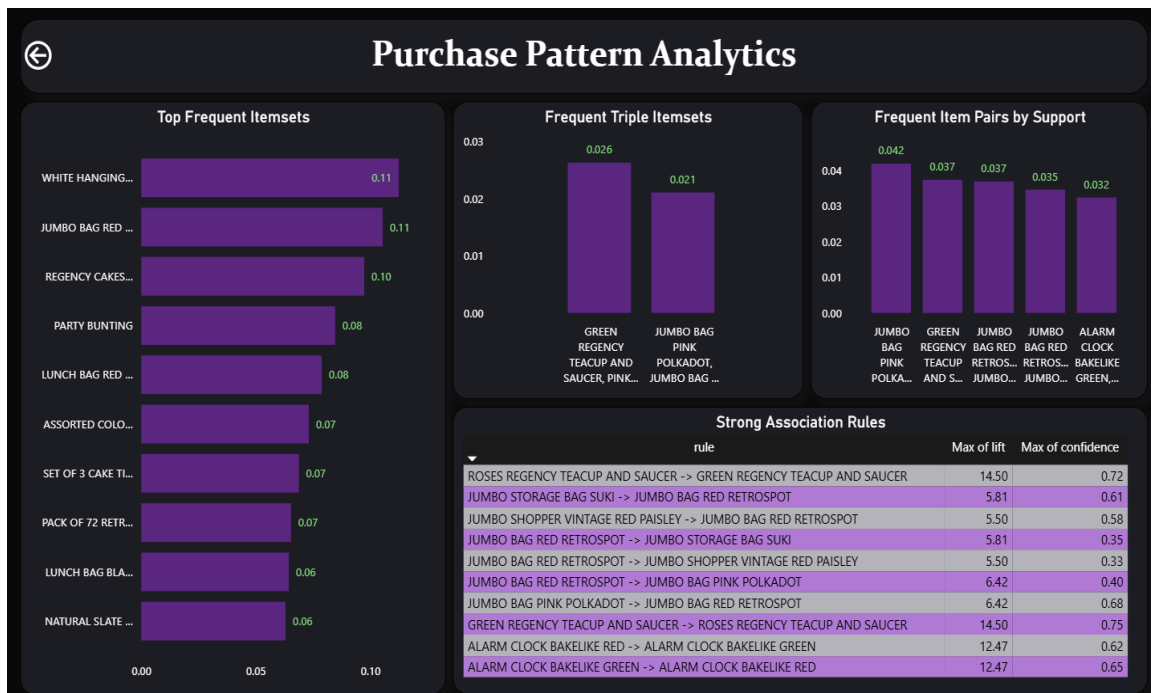


Fig: Apriori Results

12. Business Insights & Recommendations

The Market Basket Analysis identified strong relationships between products frequently purchased together, revealing consistent customer buying patterns. High-lift and high-confidence associations highlighted reliable product combinations suitable for bundling and cross-selling.

In addition, purchase trend analysis revealed periods of higher demand and consistently high-performing products, providing context for inventory and promotional planning.

Based on these findings, the following actions are recommended:

- **Bundle frequently co-purchased products** to increase average order value
- **Enable cross-selling recommendations** using strong association rules
- **Optimize product placement** for items commonly bought together
- **Prioritize inventory planning** for high-frequency and high-association products
- **Design targeted promotions** around strong product combinations

These insights support data-driven decisions that improve sales performance, inventory efficiency, and customer experience.

13. Tools and Technologies Used

- Python (Pandas, NumPy, mlxtend)
- Jupyter Notebook

- Power BI
- Market Basket Analysis (Apriori Algorithm)

14. Conclusion

This project demonstrates a complete end-to-end analytics workflow, beginning with raw data ingestion and cleaning in Python, followed by descriptive and trend-based visualization in Power BI. Market Basket Analysis using the Apriori algorithm was then applied to uncover meaningful product associations, with results visualized for clear interpretation. The insights generated provide actionable business value and highlight strong capabilities in data preparation, analytical modeling, and data visualization.

Prepared By:

S. Asmita

T. Sumanjali

Project Team ID: PTID-CDA-OCT-25-800

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